

# Boyce-Codd Normal Form (BCNF) Relational Database Design

> **Definition:** A relation is in **Boyce-Codd Normal Form (BCNF)** (also known as 3.5NF) if and only if for every non-trivial functional dependency  $X \rightarrow Y$ ,  $X$  is a strict Super Key.

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## 1. Detailed Technical Explanation

BCNF is a stricter version of 3NF that handles cases where a relation has multiple overlapping candidate keys.

### 3NF vs BCNF Comparison:

| Property | 3NF | BCNF |

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### Conversion Example to BCNF

#### Relation ADVISOR(Student\_ID, Subject, Advisor\_Name)

- \*\*A

#### Relation Status:

- Is it

#### BCNF Decomposition:

1. \*\*AD

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## 2. Core Concepts & Memory Keywords

- **Strict Super Key Rule:** Every determinant in a functional dependency must be a super key.

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## 3. Must-Write Points for Exams

- BCNF eliminates all redundancy resulting from functional dependencies.

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4. Quick Recall Flow

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3NF R